SOFTWARE ENGINEERING

OBJECT ORIENTED DESIGN EXERCISE REPORT

City Vehicle Tax Management System

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Course: Software Engineering

Exercise: Object Oriented Design / Vehicle Tax System

1. Introduction

The city needs a way to keep track of annual taxes for cars and motorcycles. Each vehicle has a plate number, maker, and model. A person is registered using a license number, first name, surname, and address. The system stores this information, manages ownership, calculates tax, and prepares annual reports.

2. System Requirements

The system allows users to add, update, delete, and search for people and vehicles. It must also record the sale of a vehicle without an owner, transfer a vehicle between registerEd people, calculate annual taxe, and produce a report of owners, vehicles, and tax totals.

2.1 Annual Tax Rules

Vehicle type	Annual tax calculation	Example
Motorcycle	EUR 0.10 x displacement (cc)	150 cc = EUR 15.00
Petrol car	EUR 1.40 x CO2 emissions (g)	120 g = EUR 168.00
Diesel car	EUR 1.80 x CO2 emissions (g)	100 g = EUR 180.00
Hybrid car	EUR 1.20 x CO2 emissions (g)	90 g = EUR 108.00

For motorcycles, the tax is defined as 10% of engine displacement. In this implementation, the calculation uses EUR 0.10 per cc as a assumption.

2.2 Ownership Rules

A person may own several vehicle or none at all. Each vehicle has at most one owner, and it can remain unowned while at a dealership. Ownership can be transferred to another registered person. The default annual report only includes vehicles with owners.

3. Use Case Diagram

Figure 1 shows the main functions available in the system. The City Tax Clerk is responsible for managing records, transferring vehicle ownership, calculating taxes, and generating annual reports

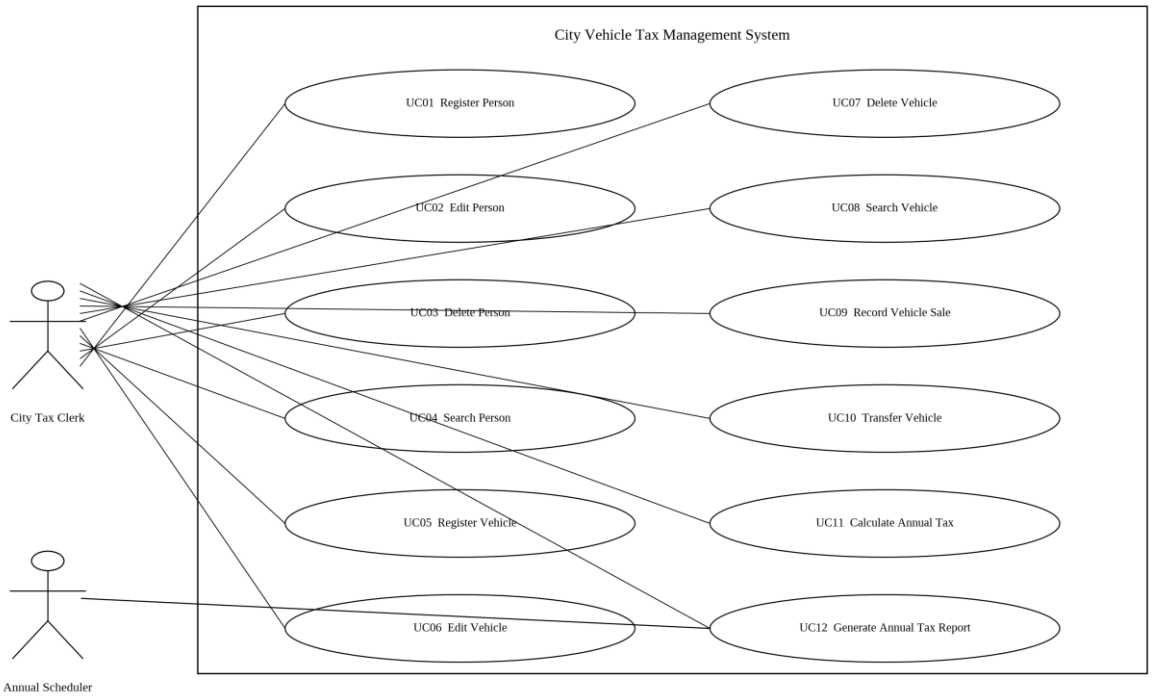


Figure 1. Use Case Diagram

4. Use Case Descriptions

These use cases explain the actions available to the clerk and the result of each action.

UC01 - Register Person. The clerk enter a license number, name, and address. The system saves the person if the details are valid and the license number is not already registered.

UC02 - Edit Person. The clerk finds a person by license number and updates their name or address. The license number does not change, and invalid details are rejected.

UC03 - Delete Person. The clerk delete a person. Their record is removed, but any vehicles they owned remain registered without an owner.

UC04 - Search Person. The clerk searches by license number, first name, surname, or address. The system displays matching records.

UC05 - Register Vehicle. The clerk enters the plate number, maker, model, vehicle type, and the relevant emission or displacement value. A valid vehicle is registered without an owner.

UC06 - Edit Vehicle. The clerk updates vehicle details or values used for tax calculations. If the vehicle category changes, the system replaces the object while keeping its plate number and owner.

UC07 - Delete Vehicle. The clerk deletes a registered vehicle. It is then removed from searches and future reports.

UC08 - Search Vehicle. The clerk searches by plate number, maker, model, or vehicle type. The system displays matching vehicles.

UC09 - Record Vehicle Sale. The clerk record the sale of an unowned vehicle to a registered person. The system assigns the buyer and calculates the annual tax. An already-owned vehicle must be transferred instead.

UC10 - Transfer Vehicle. The clerk chooses an owned vehicle and a different registered person. The system checks both records and updates the owner. Transfers are rejected if either record is missing, the vehicle has no owner, or the new owner is the current owner.

UC11 - Calculate Annual Tax. The clerk requests the annual tax for a registered vehicle. The system uses the calculation for that vehicle type and returns the amount.

UC12 - Generate Annual Tax Report. The system calculates taxes for owned vehicles, groups them by owner, and returns vehicle amounts, owner subtotals, and the grand total.

5. Class Diagram

Figure 2 shows the classes used in the design. Vehicle is an abstract class, with Car and Motorcycle as subclasses. PetrolCar, DieselCar, and HybridCar extend Car and calculate tax using their own rates. A vehicle may refer to one Person as its owner, or have no owner. VehicleTaxSystem manages the records and prepares the reports.

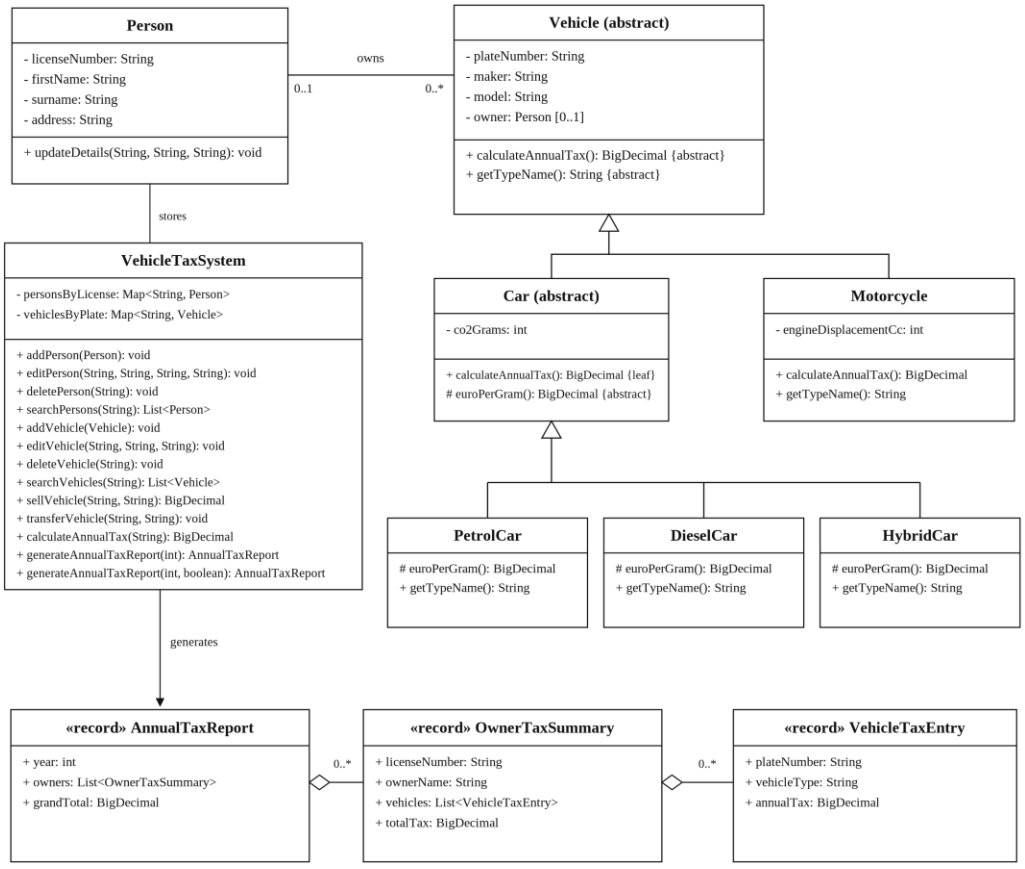


Figure 2. Class Diagram

VehicleTaxSystem stores people by license number and vehicles by plate number using maps. An AnnualTaxReport contains an owner summary for each reported owner, and each OwnerTaxSummary contains VehicleTaxEntry records. The diagram shows the main fields and methods rather than every helper method.

6. Sequence Diagram - Transfer Vehicle

The transfer operation changes the owner of a vehicle that already belongs to someone. Figure 3 shows a successful transfer.

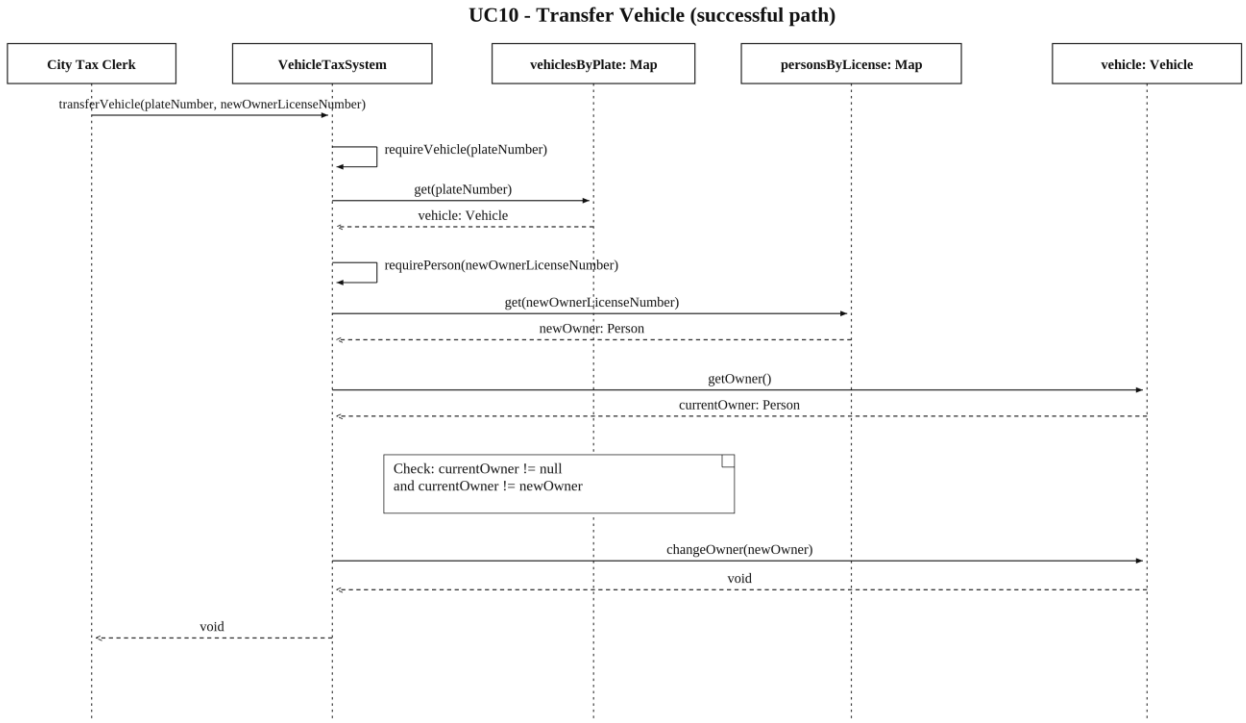


Figure 3. Transfer Vehicle Sequence Diagram

The clerk enters the vehicle plate number and the new owner's license number. VehicleTaxSystem finds both records and checks the current owner. If the vehicle is owned and the new person is different, the owner is updated. The transfer is rejected if a record is missing, the vehicle has no owner, or the selected person already owns it.

7. Sequence Diagram - Annual Taxes Listing

Figure 4 shows how the system prepares an annual report for the current vehicle owners.

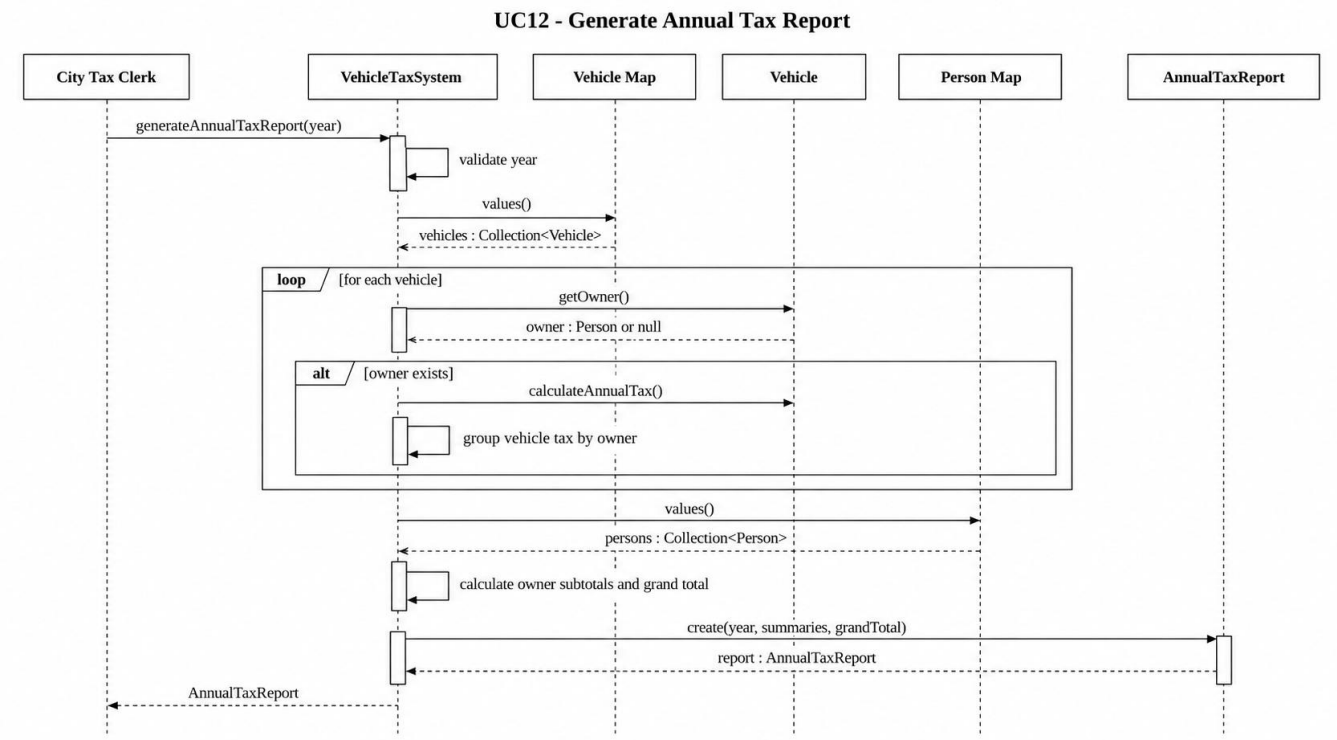


Figure 4. Annual Taxes Listing Sequence Diagram

When the report is requested, the system checks the registered vehicles and skip those without owner. It calculates the tax for each owned vehicle, groups the entries by owner, and adds the subtotals and final total. People who own no vehicles are normally left out, but the program can include them with a zero total.

8. Java Implementation

The project uses Java 17 and the standard Java library. VehicleTaxSystem keeps people and vehicles in LinkedHashMap collections. Person and Vehicle store the main details. The Car and Motorcycle classes, including the different car types, handle the tax calculations. VehicleTaxEntry, OwnerTaxSummary, and AnnualTaxReport hold the report data.

The main classes are Person, Vehicle, Motorcycle, Car, PetrolCar, DieselCar, HybridCar, VehicleTaxSystem, VehicleTaxEntry, OwnerTaxSummary, and AnnualTaxReport. Key methods include addPerson(), addVehicle(), searchPersons(), searchVehicles(), sellVehicle(), transferVehicle(), calculateAnnualTax(), and generateAnnualTaxReport().

Tax amounts use BigDecimal. The program uses polymorphism by calling calculateAnnualTax() on a Vehicle object, so each vehicle type applies its own calculation.

8.1 Running the Program

The main program starts from com.city.vehicletax.DemoApp. The tests are in com.city.vehicletax.VehicleTaxSystemTest. With Java 17 or later installed, run these commands from the project folder:

```
javac -encoding UTF-8 -d out $(find src test -name '*.java')
java -cp out com.city.vehicletax.DemoApp
java -cp out com.city.vehicletax.VehicleTaxSystemTest
```

The compilation command is written for Linux/macOS. The original project package also includes scripts for Windows and Linux/macOS.

9. Example Annual Tax Report

In this example, Alya owns a diesel car. Bima owns a petrol car that was transferred from Alya, as well as a motorcycle. The hybrid car has no owner, and Citra owns no vehicles. For that reason, the default report lists only Alya and Bima.

Owner	Vehicle	Annual tax (EUR)
Alya	CAR-002 - Diesel car (100 g CO2)	180.00
Bima	CAR-001 - Petrol car (120 g CO2)	168.00
Bima	MOTO-001 - Motorcycle (150 cc)	15.00
Alya - subtotal		180.00
Bima - subtotal		183.00
Grand total		363.00

10. Testing

The program was compiled and tested usin VehicleTaxSystemTest, which passed 47 automated checks. These checks cover tax calculations, sales, transfers, unowned vehicles, searching, updates, validation, and report totals.

11. Assumptions and Limitations

This design uses unique, fixed license and plate numbers. If a person is deleted, their vehicles remain in the system without an owner. Reports are based on current ownership, not previous transfers. The project does not include ownership history, prorated tax, a database, or a graphical interface.

12. Conclusion

The project covers the main tasks: managing people and vehicles, recording sales and transfers, calculating annual taxes, and listing totals by owner. The UML diagrams show the system design, and the Java classes implement these functions.